

Beyond the Camera

Week 8 — Tuesday Lecture

Your drone map is perfect. Which question can it answer?

A) Where is the ground under those trees? · B) Is the roof insulation wet? · C) Is this lawn about to turn brown? · D) How big is this stockpile?

ANSWER Only D

- **A** needs to reach the ground through the canopy — **LiDAR**
- **B** needs heat, not colour — **thermal**
- **C** needs light the eye can't see — **multispectral**
- Today: one sensor for each question an RGB camera can't answer, and what fools each one

Today

- Three sensors, one question each
- What fools each one
- How processing changes with the sensor
- The other sensors drones carry
- Pick the sensor

Figure 3: What a drone can carry

Most small drones have one fixed camera. Larger aircraft take interchangeable payloads, so one airframe serves many jobs.



Standard camera

color photos and video, what your eye sees



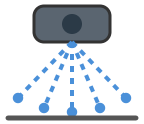
Thermal

surface temperature, shown as an image



Multispectral

light beyond visible, such as near infrared



LiDAR

distance, by timing laser pulses



Gas detector

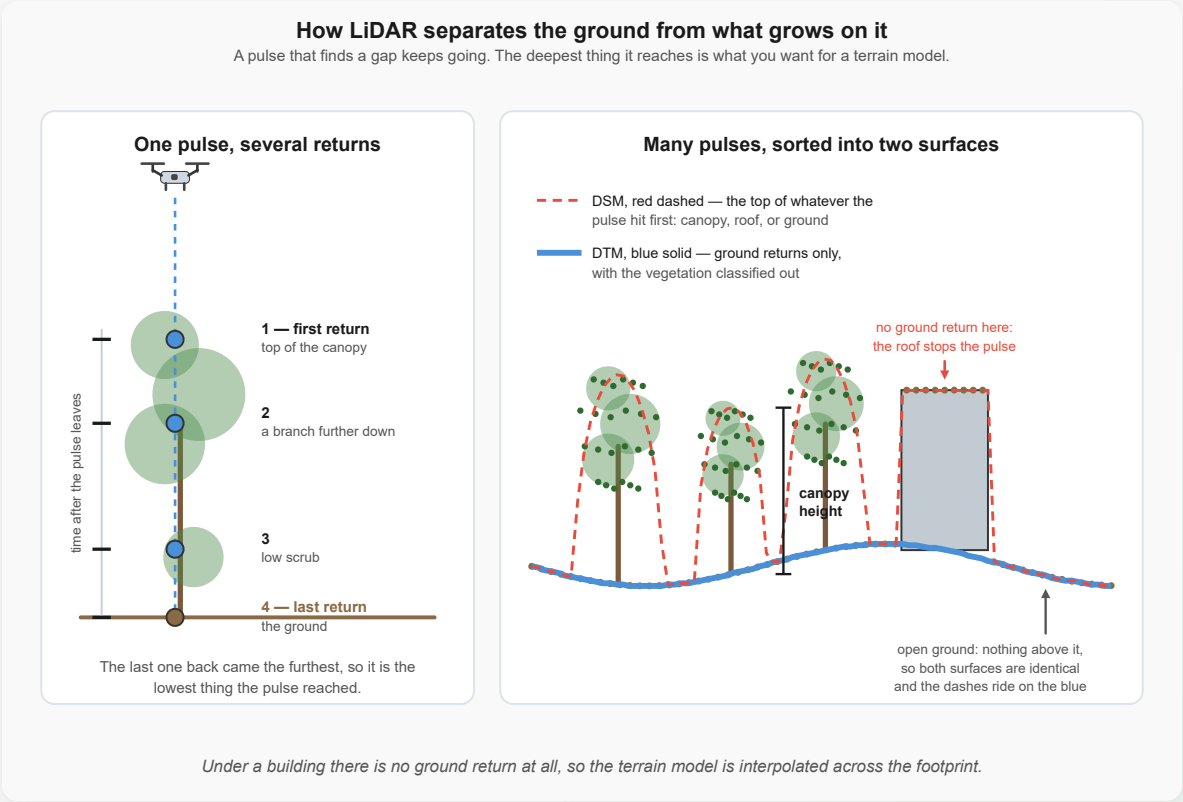
methane, CO and other gases in the air

The same aircraft can carry a different instrument. Which one depends on the question, not the drone.

Can a drone LiDAR map the ground under a tree?

A) No · B) Yes — the laser goes through the leaves · C) Yes — through gaps in the leaves

ANSWER C — through the gaps



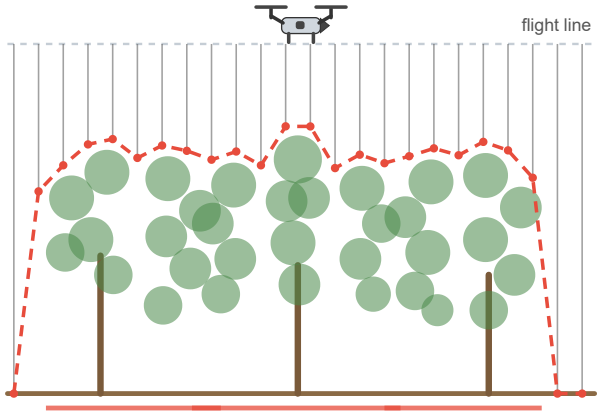
A leaf stops the pulse. Pulses that find a gap come back from the ground: one pulse, several returns.

Photos see the canopy; pulses reach the ground through the gaps

A leaf stops a laser pulse just as it blocks a camera's view. The difference is the holes between leaves: a pulse that finds one keeps going.

Photographs: the surface is the canopy top

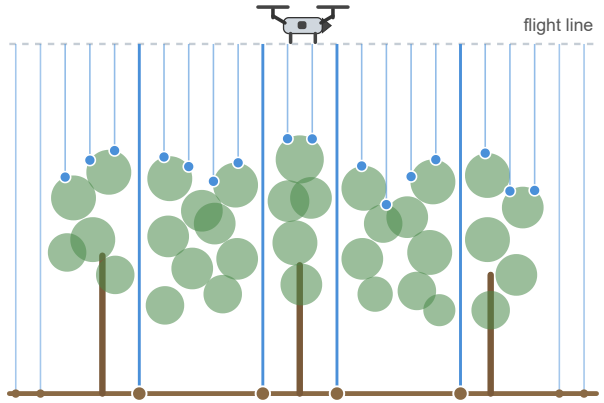
the surface the photos reconstruct
every camera ray stops at the first leaf



red: ground that no photo ever saw
the model there is a guess copied from the canopy

LiDAR: some pulses find a gap

stopped by a leaf: a canopy return
through a gap: a ground return



here: 20 pulses over the crowns, 4 reached the ground
a leaf is not transparent; the pulse went between leaves

Photographs: 0 ground points under the crowns
the ground has to be seen in two photos to be modelled

20 % gaps × 200 pulses/m² ≈ 40 ground points/m²
plenty for a terrain model on a 1 m grid

Dense evergreen canopy leaves few gaps; bare winter branches leave many, so a deciduous site maps best with the leaves off.

Photos see the top of the canopy. Laser pulses reach the ground wherever there is a gap.

200 pulses per m² · 20 % gaps

A canopy lets 20 % of the pulses through. How many ground points per m²? Is that enough for a 1 m DTM? *(Illustrative numbers.)*

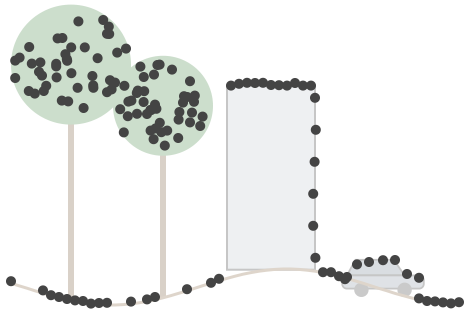
ANSWER About 40 per m² — plenty

- 200 × 0.20 = **40** ground points in every square metre: a 1 m grid, even 0.5 m, is fine
- Dense evergreen, 2 % gaps: about **4** per m² — usable, but rough
- Photos under the same canopy: **zero** ground points
- Fly in winter, when leaves are off, and the gaps get bigger

From points to a bare-earth DTM

Every point gets a label, and only the ground points build the terrain model. Where cover was dense, the surface between them is interpolated and less certain.

1. All the points

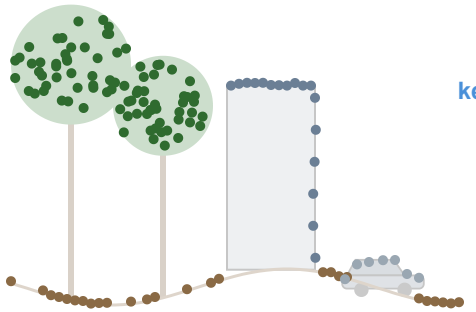


a slice through the cloud: canopy,
a roof, a parked car, the ground

classify →

2. Classified

- ground
- vegetation
- building
- other: the car

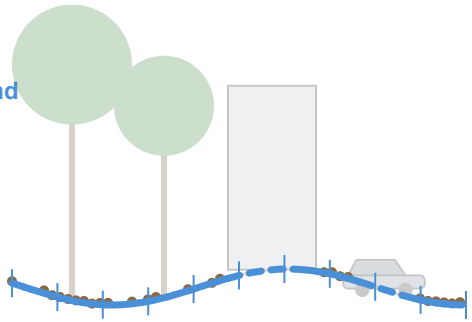


a filter sorts them by shape:
the lowest smooth surface is ground

keep ground →

3. Ground only, gridded

- DTM through the ground points
- interpolated: no points under it



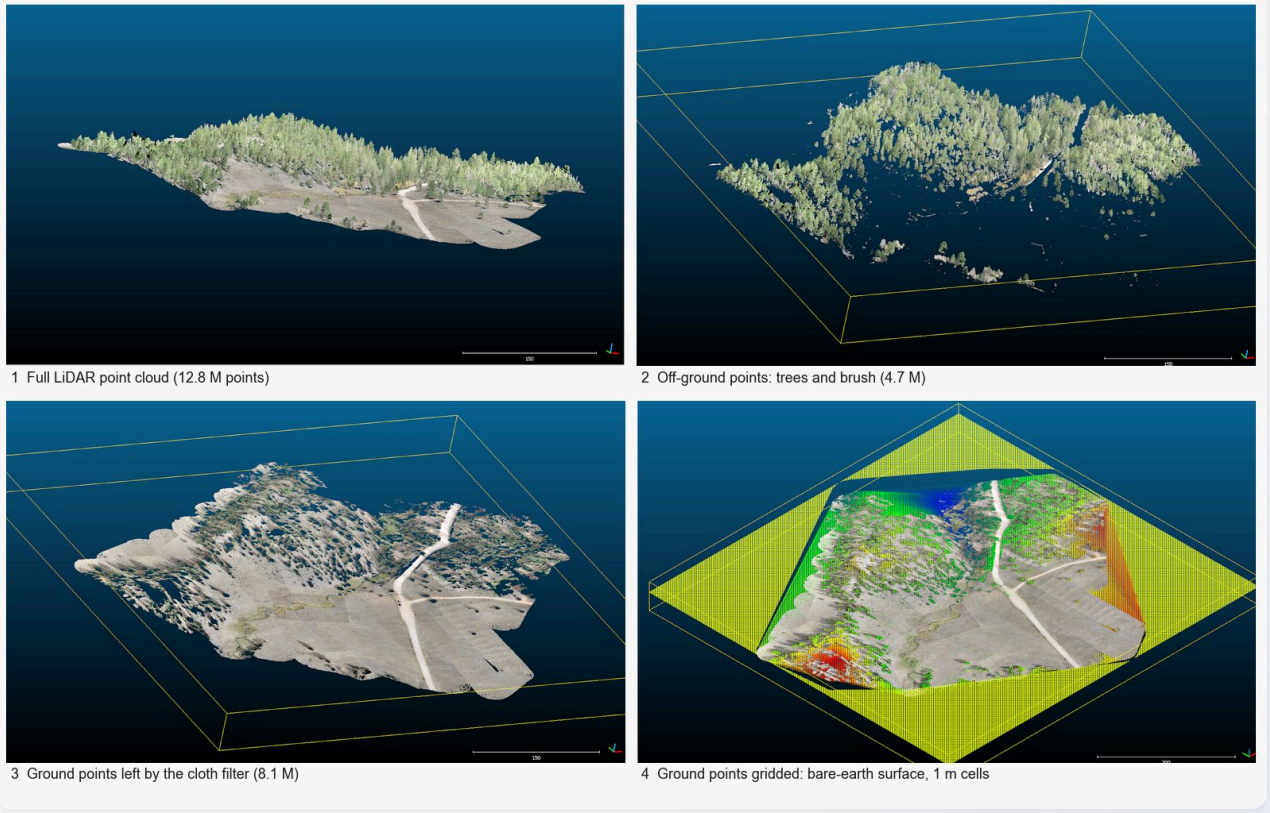
grid it: one height per cell, the
ticks; the trees and roof are gone

Software does the sorting with filters such as a cloth simulation filter. Check its answer where the cover was dense or the ground steep.

Processing LiDAR: label every point, keep the ground, then grid it into a bare-earth DTM.

Our own LiDAR: removing the vegetation

From point cloud to bare earth: CloudCompare cloth simulation filter



1 Full LiDAR point cloud (12.8 M points)

2 Off-ground points: trees and brush (4.7 M)

3 Ground points left by the cloth filter (8.1 M)

4 Ground points gridded: bare-earth surface, 1 m cells

A BYU LiDAR flight processed with a cloth-simulation filter: full cloud, ground separated from vegetation, bare-earth surface.

You route stormwater across a brushy slope using the DSM.

Where does the water go wrong?

ANSWER Every bush becomes a hill

- To a **DSM**, a shrub is a mound: flow bends around crowns and ponds behind hedges
- A bridge or culvert in a DSM is a **dam**: the model floods upstream
- Water runs on the **ground** — use the **DTM**
- A DSM gives *confidently wrong* answers here

Can a thermal camera see a leaking pipe inside a wall?

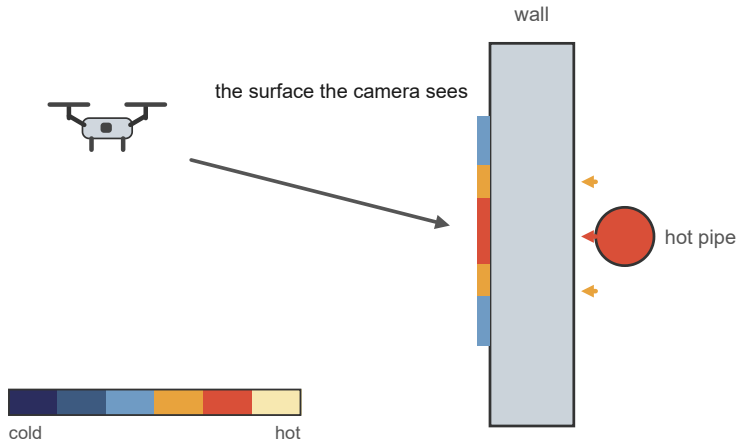
A) Yes · B) No · C) Only if the leak changes the wall's surface temperature

ANSWER C — it reads surfaces

What a thermal camera is actually reading

It measures the temperature of the surface facing it. Anything hidden shows up only if it warms that surface.

A warm pipe behind a wall



The pipe is invisible. The warm stripe it leaves on the wall face is not.

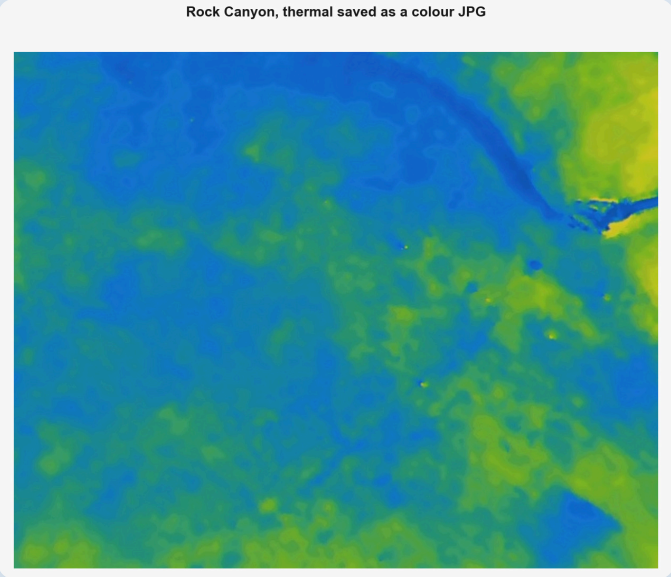
What it cannot do

- **See through a wall**
  it reads the wall's face
- **See through glass**
  glass is a mirror at these wavelengths
- **See under water**
  the surface is all it gets
- **Trust a shiny surface**
  polished metal reflects, see below

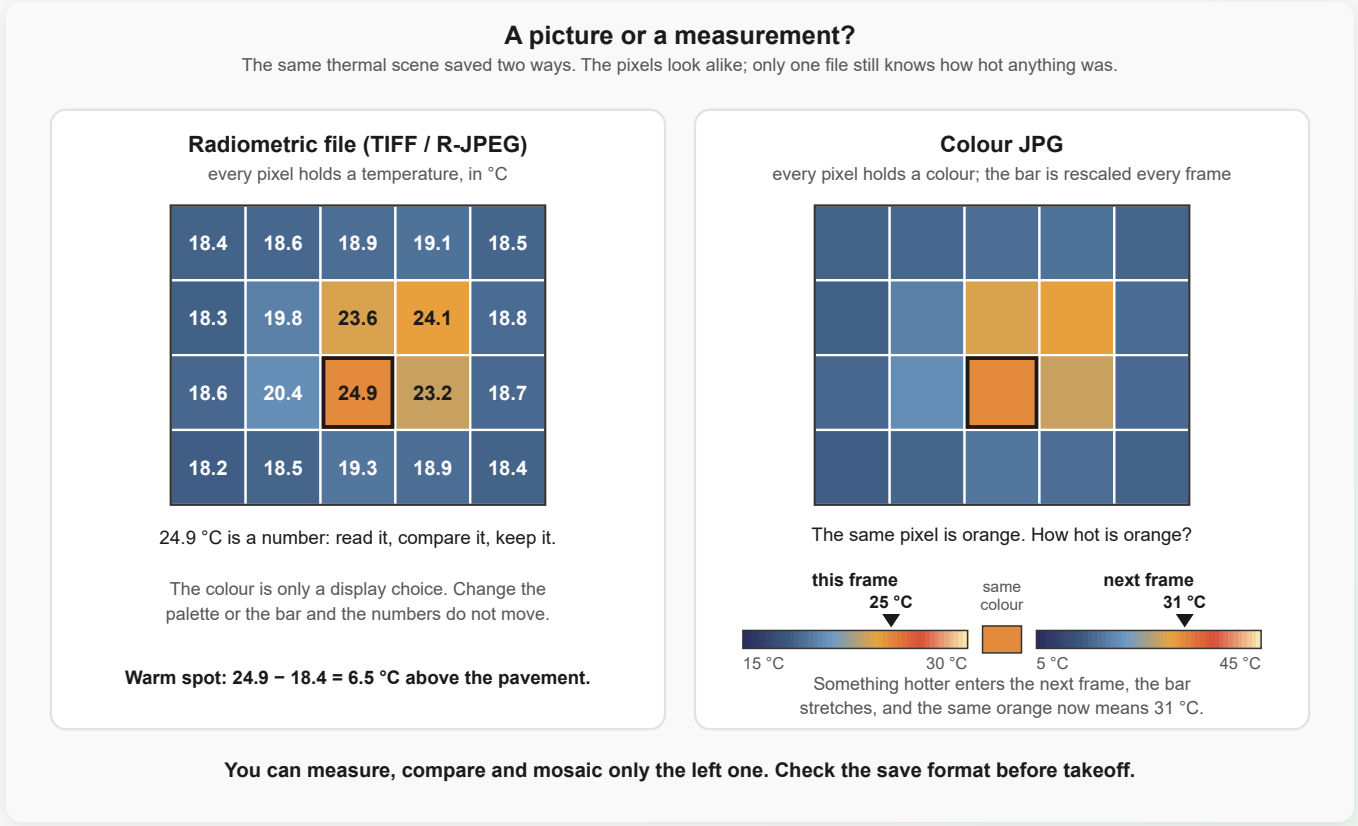
A thermal camera measures the temperature of the surface it sees. You work backward from the pattern to the cause.

Saved as a colour JPG

Our Rock Canyon thermal. How hot was the warmest spot? A) Estimate it from the colours · B) We can't tell



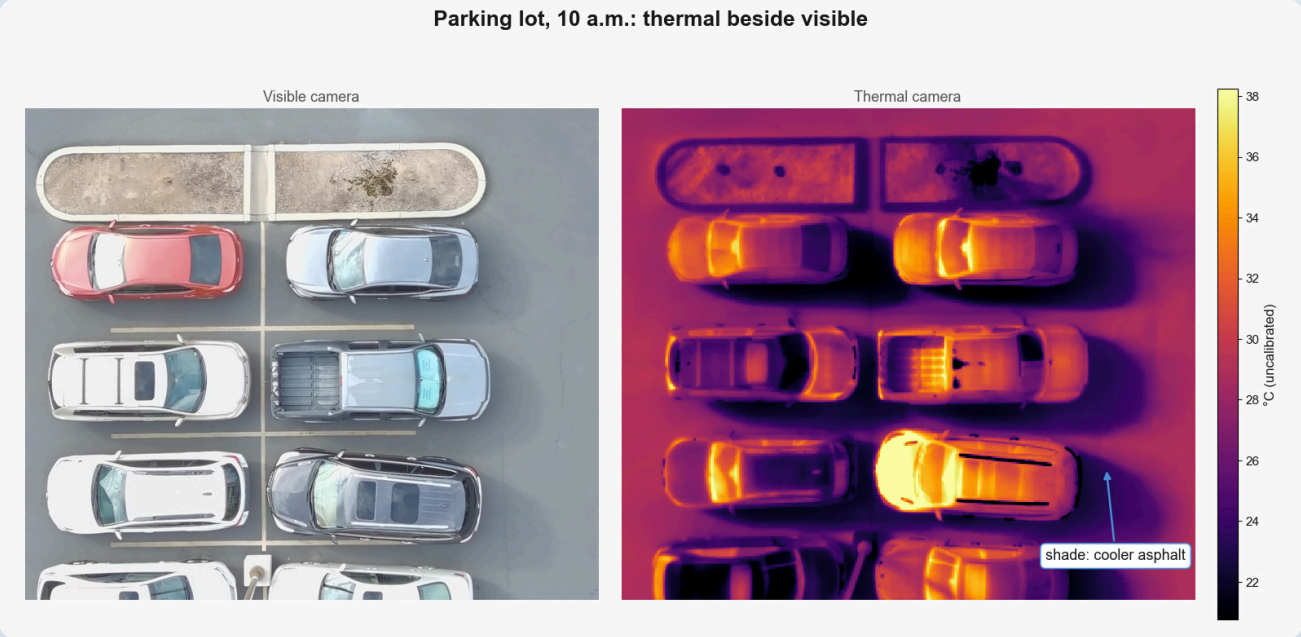
ANSWER B — a picture is not a measurement



A radiometric file stores a temperature in every pixel. A colour JPG stores a colour, rescaled every frame.

Dark shapes beside every car

Parking lot, 10 a.m. What are they, and why are they cooler?



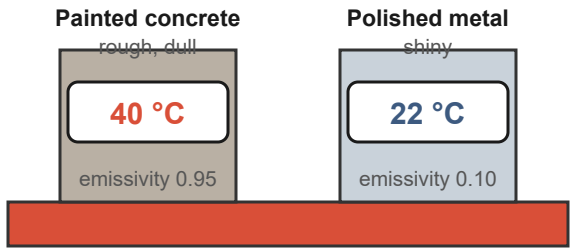
ANSWER Shade

- Sunlit asphalt has been warming since sunrise; shaded asphalt has not
- A thermal image shows **heat history** — minutes to hours — not just this moment
- A shadow that has just moved still reads cool for a while
- On a bridge deck, a patch like this could be mistaken for a defect

Why a thermal image can lie to you

The camera measures radiation and converts it to a temperature. How well a surface radiates changes the answer.

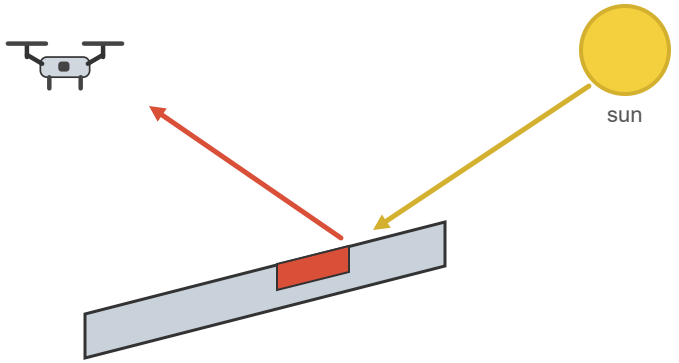
Same temperature, two readings



one hot plate, both blocks at 40 °C

The metal is exactly as hot. It just radiates badly, so the camera reads it far too cold.

A hotspot that is not hot



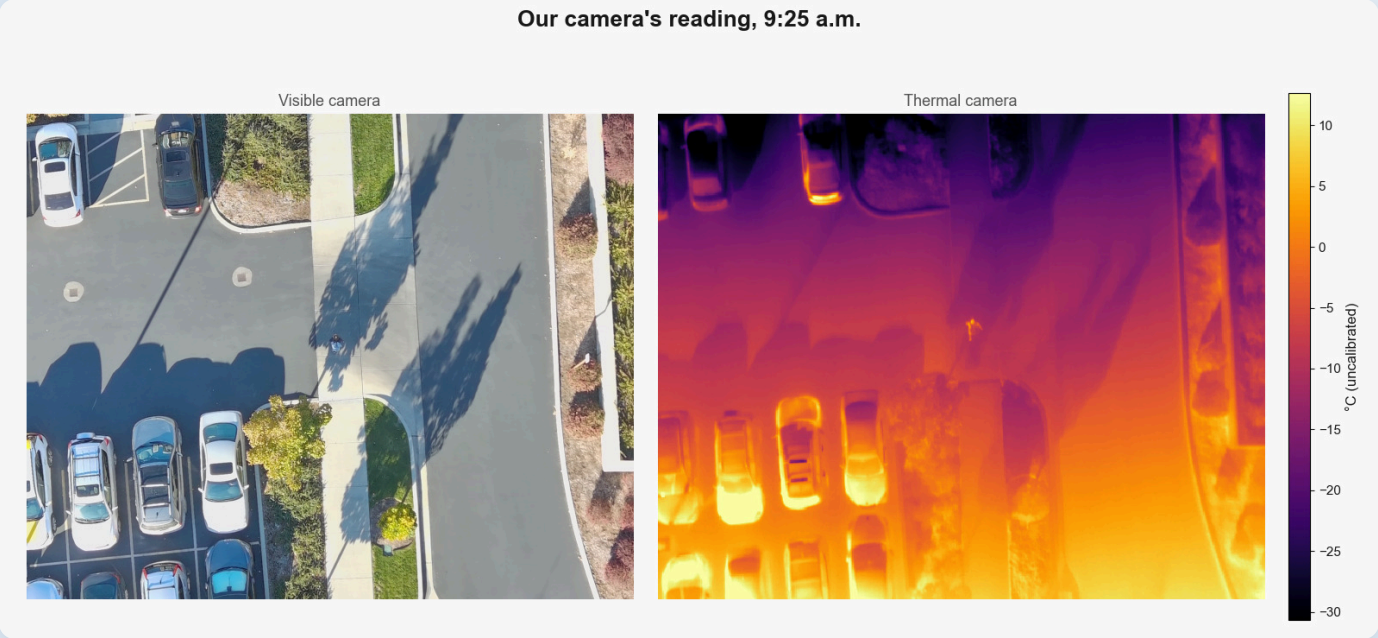
bare metal roof

The camera sees reflected sky and sun, not the roof. Check every hotspot against the ordinary photo.

Shiny surfaces reflect their surroundings instead of glowing on their own. The camera reads both.

Sunlit asphalt at −8 °C?

Our camera's reading at 9:25 a.m. Do you believe it? What would you check?

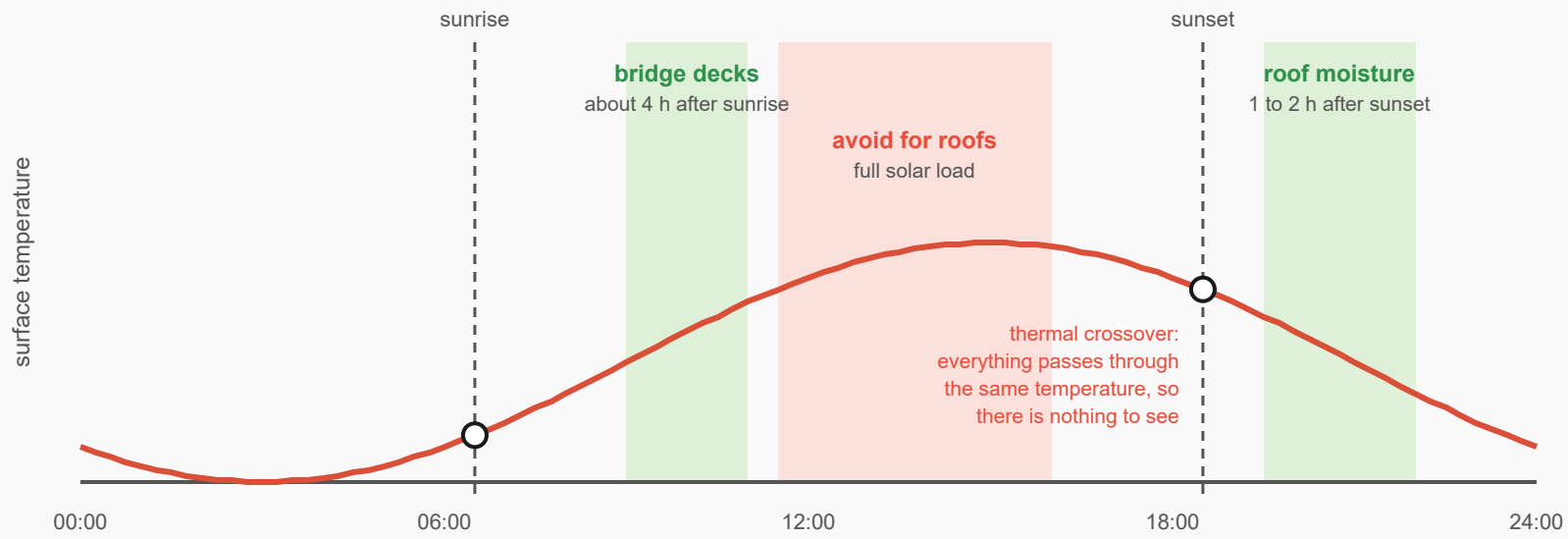


ANSWER Not until it's checked

- Put something of **known temperature** in the frame: a water bucket, a contact thermometer
- Give the camera time to **warm up** before the flight
- Check the **emissivity** setting and the **viewing angle**
- Check how the file stores temperature
- A number is not a measurement until it has been checked

When to fly a thermal survey

Thermal contrast comes from things heating and cooling at different rates. Fly when that difference is largest.



Wet insulation holds heat, so it glows after sunset. A delaminated bridge deck heats faster than sound concrete, so it shows in the morning.

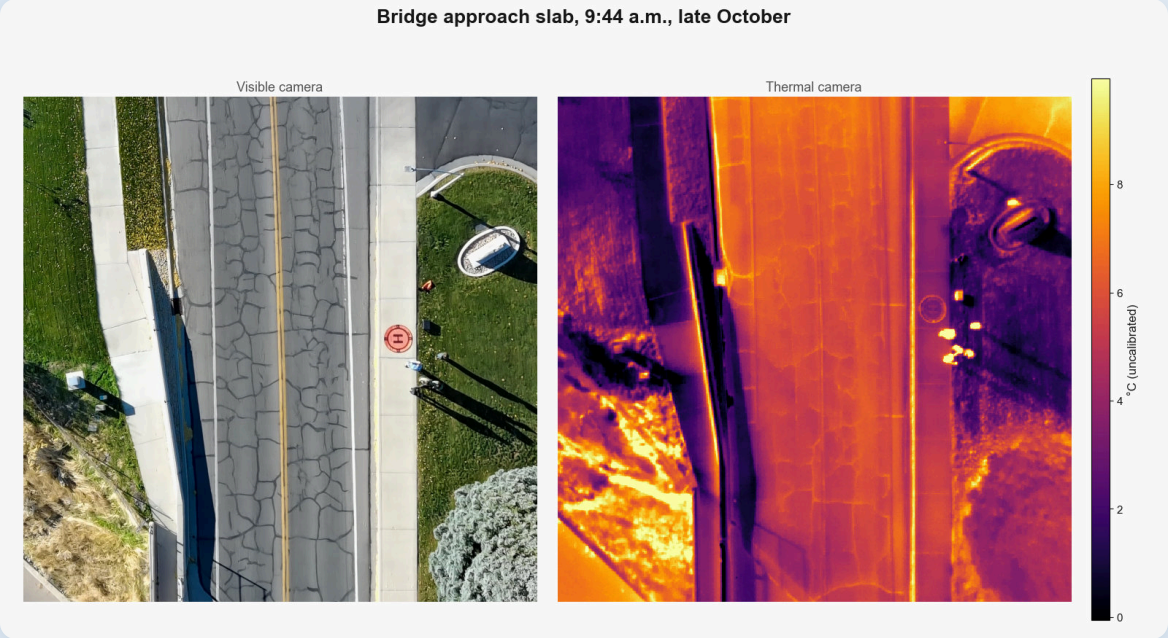
Near sunrise and sunset everything passes through the same temperature. Fly then and you will see nothing at all.

Wait at least 24 hours after rain, and skip the survey if it is windy.

When you fly decides what you can see.

Cracks show. Is it delamination?

Bridge approach, 9:44 a.m., late October. Can we report delamination? When should we have flown?

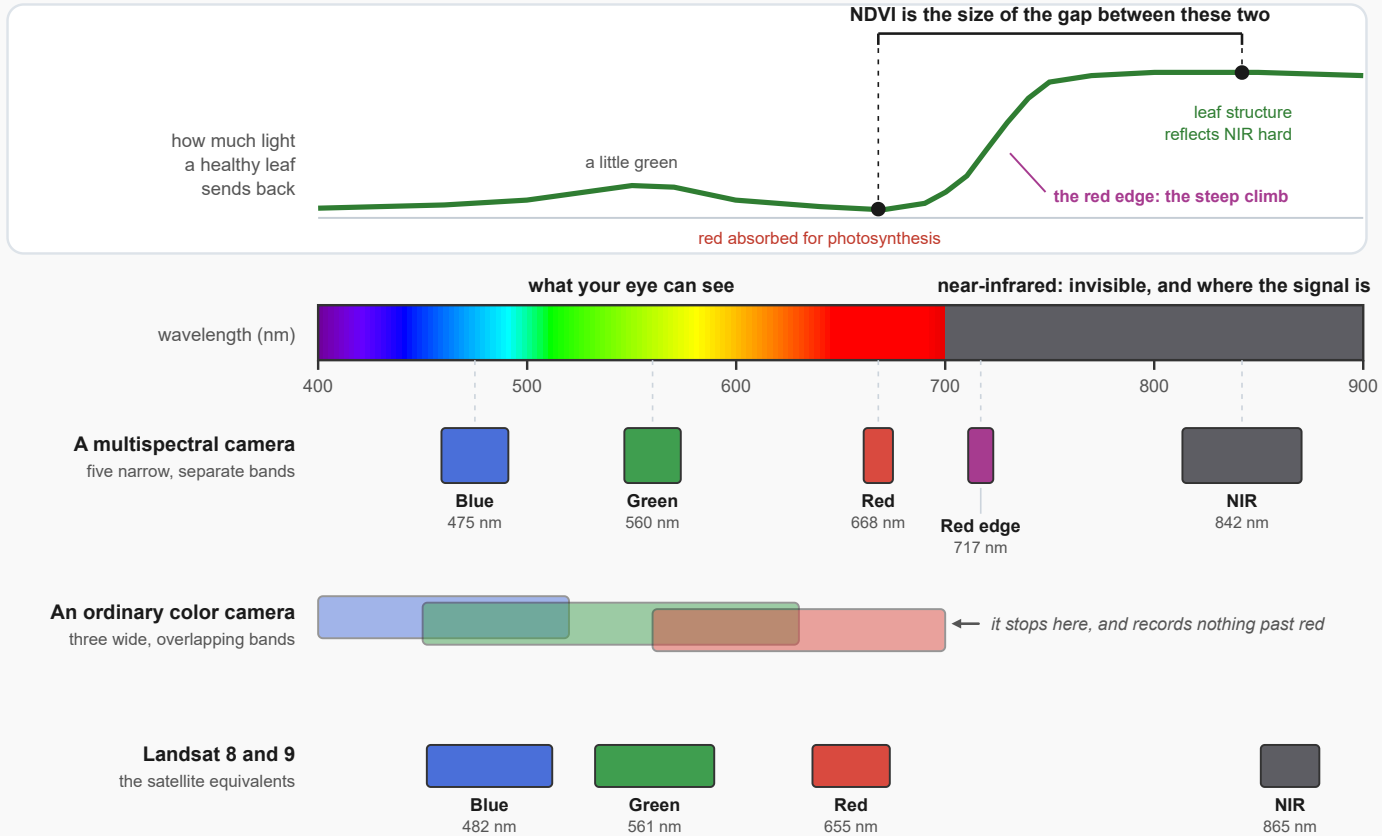


ANSWER Cracks, yes. Delamination, not yet.

- Cracks are on the surface, and the RGB photo shows them too
- Delamination is a hidden void that shows only while **heat is flowing** into the deck
- Under two hours after sunrise, with a low autumn sun: too little heat had gone in
- Fly several hours into a sunny day, on a dry deck with low wind
- Report: *cracks mapped, delamination inconclusive, refly*

Where the multispectral bands sit on the spectrum

Centers and widths vary between cameras. These are typical of a five-band survey camera.



Landsat also carries two shortwave-infrared bands at 1610 and 2200 nm, far off the right of this chart, that no drone camera has.

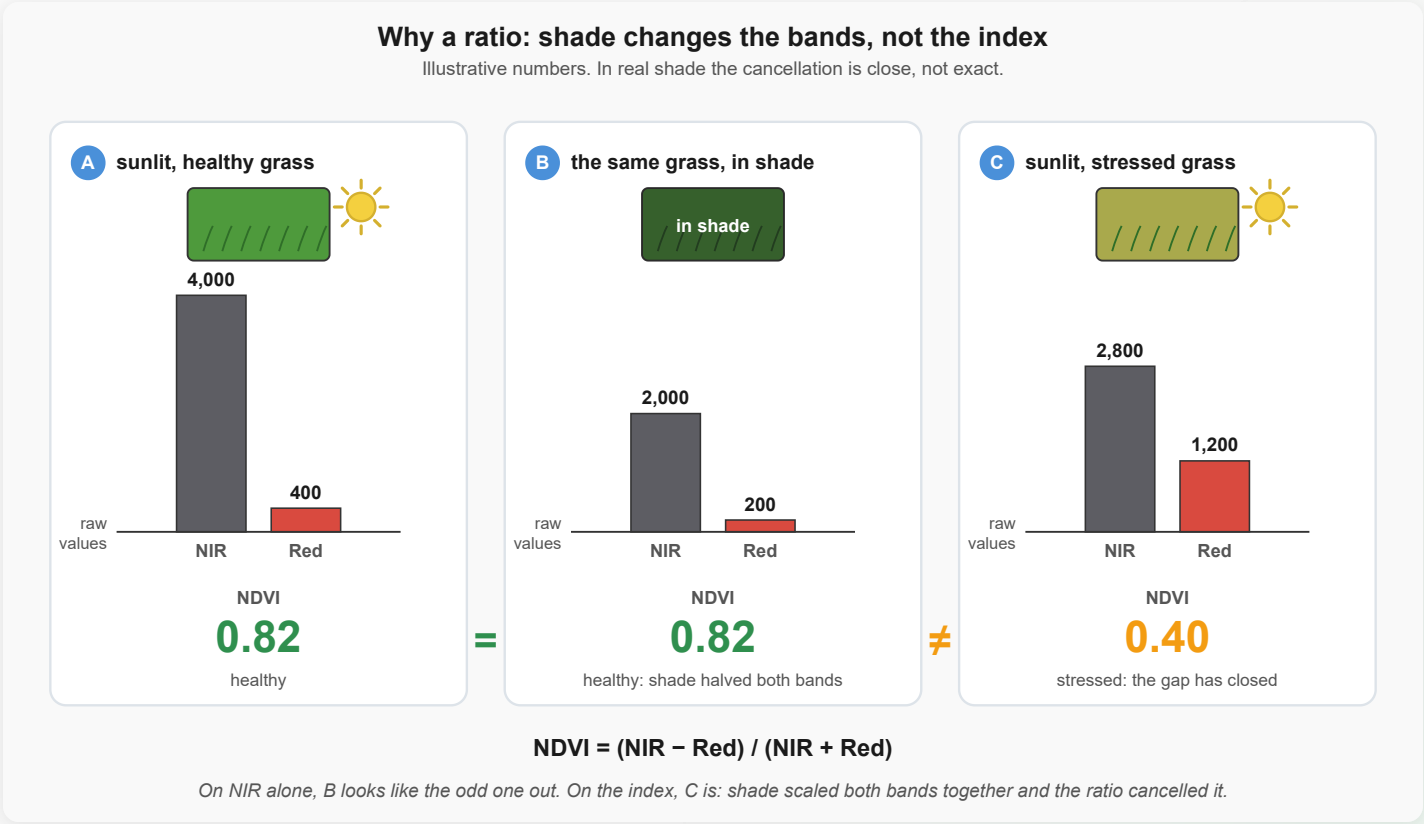
Healthy leaves absorb red light and reflect near-infrared strongly.

Three patches of the same lawn. Which one is stressed?

A, sunlit: NIR 4000, Red 400 · B, shaded: NIR 2000, Red 200 · C, sunlit: NIR 2800, Red 1200

$$\mathrm{NDVI} = \frac{\mathrm{NIR} - \mathrm{Red}}{\mathrm{NIR} + \mathrm{Red}}$$

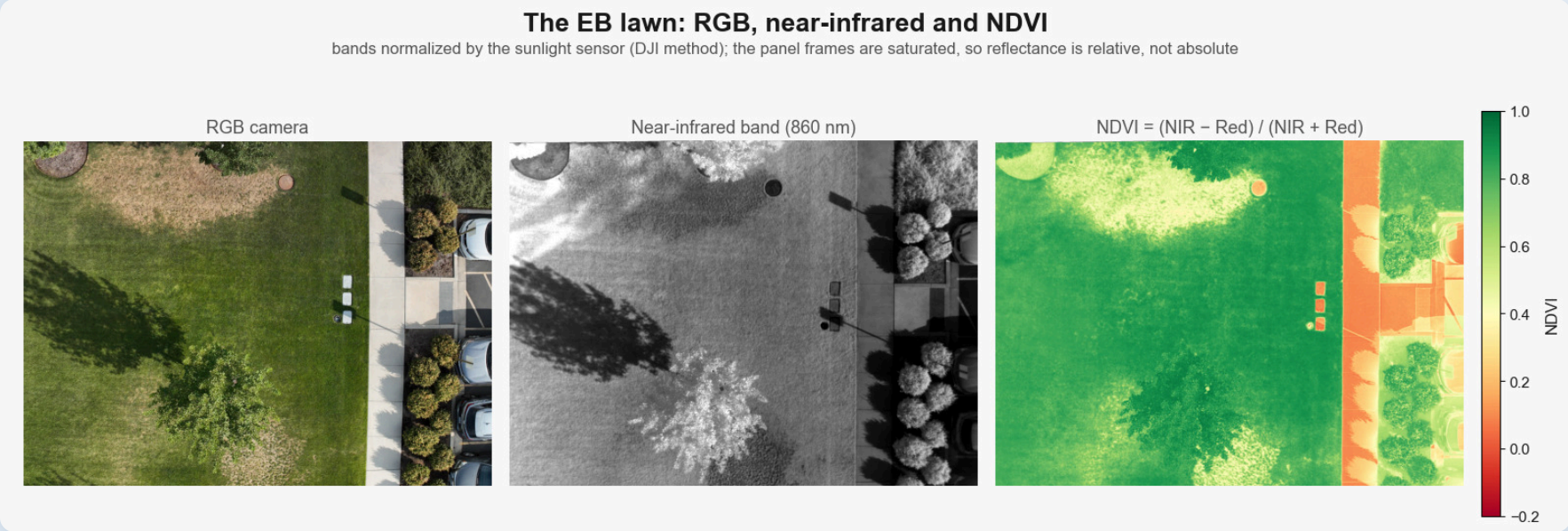
ANSWER C — the shade cancels out



NDVI: A 0.82, B 0.82, C 0.40. Shade halved both bands, so the ratio did not change.

Low spots on the NDVI

Our map of the lawn east of the EB. Does a low spot prove it needs water?



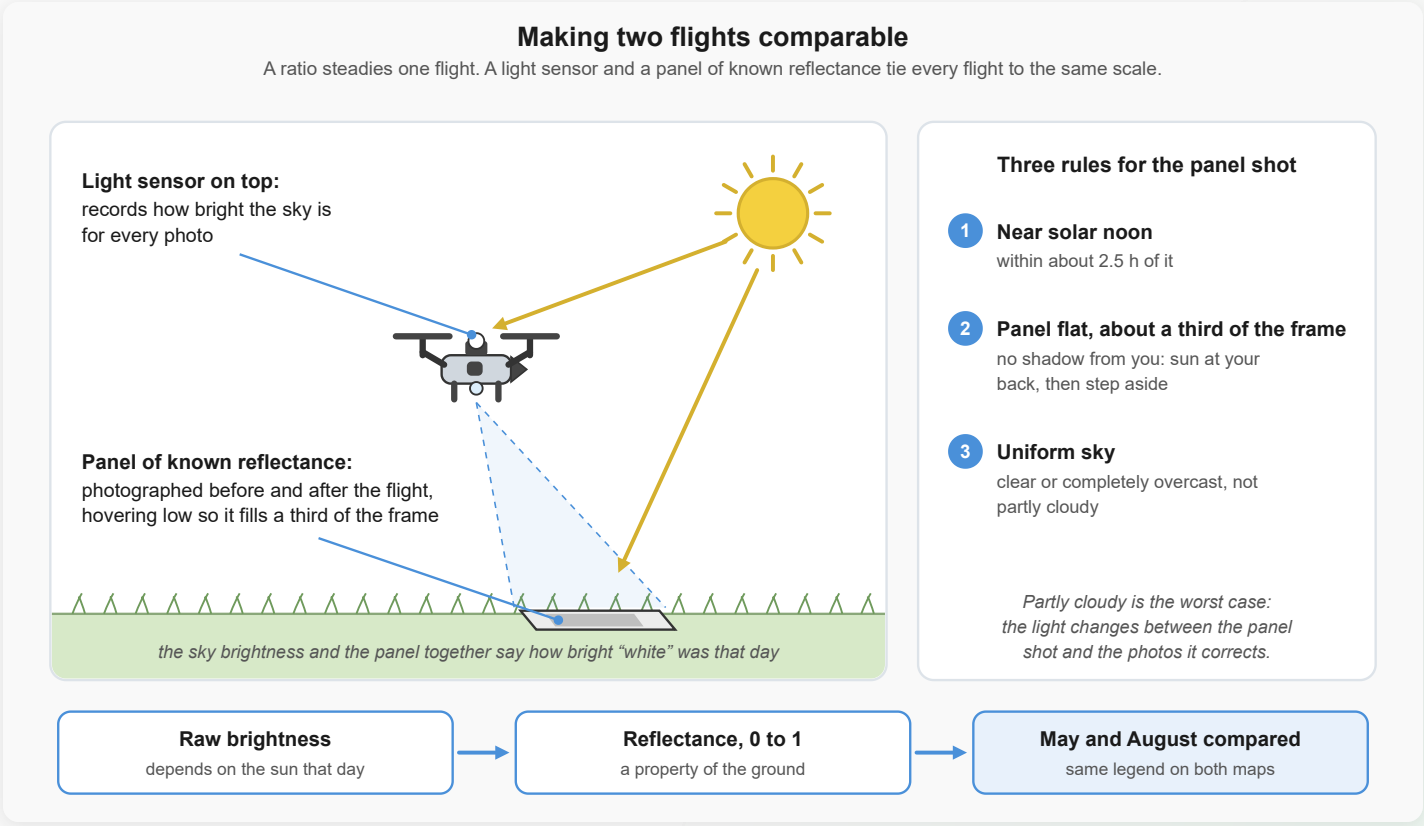
ANSWER It shows less green leaf — not why

- Low NDVI means less healthy leaf here; it does not say what caused it
- Sprinkler gaps, compacted soil, disease, bare soil and fresh mowing all lower it
- Arcs that line up with sprinkler heads point at irrigation
- Walk to one spot and look before you spend money

Same lawn: NDVI 0.70 in May, 0.60 in August.

Did it get worse? A) Yes · B) We can't tell yet

ANSWER B — not without calibration



A ratio steadies one flight. A light sensor and a reflectance panel make two flights comparable.

Same flight plan, different last step

Sensor	What processing adds	You get
RGB	The Week 7 pipeline	Orthophoto, DSM, mesh
LiDAR	Classify every point; keep the ground	DTM under vegetation
Thermal	Keep the radiometric file; set emissivity	Surface temperatures
Multispectral	Light sensor and panel; reflectance; index	NDVI you can compare

Other sensors drones carry

Sensor	Measures	Civil use	On drones today
Ground-penetrating radar	Radar echoes from buried layers	Utilities, voids, pavement layers	Specialised; soil limits depth
Imaging radar (InSAR)	Millimetres of ground movement	Subsidence, dam and levee movement	Satellite, not drone
Magnetometer	Distortion of Earth's magnetic field	Buried steel pipe, tanks, old wells	Specialised, well proven
Gravimeter	Tiny changes in gravity	Large density contrasts, geology	Experimental; too coarse for small voids
Methane laser, gas camera	Gas absorbing laser or infrared light	Landfill, pipeline, wastewater leaks	Laser routine; camera specialised
Hyperspectral camera	Hundreds of narrow colour bands	Minerals, water quality, plant species	Specialised; heavy data
Green LiDAR, sonar	Water depth	River and reservoir depth, scour	Specialised; clear water for LiDAR
Air-quality sensors	Particles and gases in the air	Dust and smoke plumes	Experimental; rotor wash skews it
Gamma spectrometer	Radioactivity	Mine tailings, contaminated land	Specialised

Is there an old well under this field?

A field becoming a park may hide an unrecorded oil well and a buried steel tank. The soil is wet clay. A teammate says "fly ground-penetrating radar." What would you fly instead, and what could it miss?

ANSWER A magnetometer

- Steel well casing and steel tanks bend Earth's magnetic field strongly
- Wet clay is where ground-penetrating radar does worst
- It misses anything not made of steel: a wood-cased old well, a fibreglass tank, plastic pipe
- Confirm with a records search and test pits before grading

Pick the sensor — and say what would fool it

Groups of 3 · 5 minutes

1. Volume of a gravel stockpile
2. Ground surface under scrub on a hillside
3. Delamination in a bridge deck
4. Vegetation cover on a reseeded construction slope (stormwater permit)

ANSWER Answers

1. **RGB photogrammetry** — the honest cheap case; fooled by a wet or shiny pile and by missing ground control
2. **LiDAR** — gaps reach the ground; fooled by dense evergreen cover
3. **Thermal, timed right** — fooled by shade, patches and reflections; confirm with a chain drag
4. **Multispectral or RGB** — RGB is often enough to count cover; fooled by comparing uncalibrated flights

What to do Thursday

- **Data Processing lab:** Data Processing
- Bring the photos your team flew at Rock Canyon
- Lab slides: Bentley iTwin Capture Modeler
- Readings for reference: LiDAR · Thermal · Multispectral